Summary in Graph

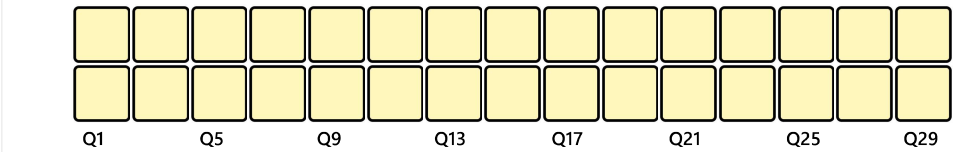
# Exam Summary (GO Classes DA Test Series 2026 | Probability and Statistics | Subject Wise Test 3)

| | | | |
|---|---|---|---|
| **Qs. Attempted:** | **0** (0 + 0) | **Correct Marks:** | **0** (0 + 0) |
| **Correct Attempts:** | **0** (0 + 0) | **Penalty Marks:** | **0** (0 + 0) |
| **Incorrect Attempts:** | **0** (0 + 0) | **Resultant Marks:** | **0** (0 + 0) |

| | |
|---|---|
| **Total Questions:** | **30** (20 + 10) |
| **Total Marks:** | **40** (20 + 20) |
| **Exam Duration:** | **90 Minutes** |
| **Time Taken:** | **0 Minutes** |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (29) out of 287 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ be the total number of text messages a random college student at Big State University receives in a year. Suppose the probability mass function (pmf) of $X$ is given by:

$$p_X(n) = \frac{c}{2^n}, \quad \text{for } n = 0, 1, 2, 3, \ldots$$

What must the value of $c$ be for $p_X(n)$ to be a valid pmf?

A. $\frac{1}{4}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. $1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $Z_1, Z_2, \ldots, Z_n \sim \mathcal{N}\left(\mu, \sigma^2\right)$ be independent and identically distributed random variables. Define

$$T = \sum_{i=1}^{n} Z_i^2$$

What is the expected value of $T$ ?

A. $\mathbb{E}[T] = n\sigma^2 + n\mu^2$

B. $\mathbb{E}[T] = \sigma^2 + n\mu^2$

C. $\mathbb{E}[T] = n\left(\sigma^2 + \mu\right)^2$

D. $\mathbb{E}[T] = n\left(\mu^2 + \sigma\right)^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If $X$ is a Binomial $(30, 0.5)$ random variable, then

A. $P(X > 15) = 0.5$

B. $P(X < 15) = 0.5$

C. $P(X > 15) > 0.5$

D. $P(X < 15) < 0.5$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ be a continuous random variable with the probability density function symmetric about $0$ . If $V(X) < \infty$, then which of the following statements is true?

A. $E(|X|) = E(X)$

B. $V(|X|) = V(X)$

C. $V(|X|) < V(X)$

D. $V(|X|) > V(X)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Let $X$ and $Y$ be random variables such that:

- $E(X) = 75$
- $E(Y) = 25$
- $\mathrm{Var}(X) = 36$
- $\mathrm{Var}(Y) = 16$
- $\mathrm{Corr}(X, Y) = 0.25$

Define $U = X + Y$. Then the value of $\mathrm{Var}(U)$ is:

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $F(x)$ be a cumulative distribution function defined as:

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1 - e^{-x}, & \text{if } 0 \le x < 1 \\ c, & \text{if } 1 \le x < 2 \\ 1 - e^{-x}, & \text{if } x \ge 2 \end{cases}$$

Which of the following describes all possible values of the constant $c$ for which $F(x)$ is a valid cumulative distribution function?
Hint: $F(x)$ must be a non-decreasing function.

A. $c \in \left[1 - e^{-2}, 1\right]$

B. $c \in \left[0, 1 - e^{-2}\right]$

C. $c \in \left[1 - e^{-1}, 1 - e^{-2}\right]$

D. $c \in \left[1 - e^{-1}, 1\right]$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The cumulative distribution function of a random variable $X$ is given by

$$F(x)=\begin{cases}0, & \text{if } x<0\\ \frac{1}{4}+\frac{1}{6}\left(4x-x^2\right), & \text{if } 0\le x<1\\ 1, & \text{if } x\ge 1\end{cases}$$

Then

$$P(X=0 \mid 0\le X<1)=$$

A. $\frac{2}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{1}{3}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ and $Y$ be two independent random variables. Which one of the relations between expectation (E), variance (Var) and covariance (Cov) given below is FALSE?

A. $E(XY)=E(X)E(Y)$

B. $\mathrm{Cov}(X,Y)=0$

C. $\mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)$

D. $E\left(X^2Y^2\right)=(E(X))^2(E(Y))^2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

In the following table, $x$ is a discrete random variable and $p(x)$ is the probability density. The standard deviation of $x$ is

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $p$(x) | 0.3 | 0.6 | 0.1 |

A. $0.18$

B. $0.36$

C. $0.54$

D. $0.6$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Probability

The probability density function of the random vector ( $X, Y$ ) is given by

$$f_{X,Y}(x,y) = \begin{cases} c, & 0 < x < y < 1 \\ 0, & \text{otherwise .} \end{cases}$$

Then the value of $c$ is equal to.....

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ be an exponential random variable such that $E\left(X^2\right) = \frac{1}{2}$.
For $x \geq 0$, which of the following is the probability density function (pdf) of $X$ ?

A. $f_X(x) = e^{-x}$

B. $f_X(x) = 2e^{-2x}$

C. $f_X(x) = 4e^{-2x}$

D. $f_X(x) = \frac{1}{2}e^{-x/2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A random variable $X$ has passion distribution. If

$$2P(X = 2) = P(X = 1) + 2P(X = 0)$$

then the variance of $X$ is

A. $\frac{3}{2}$

B. $2$

C. $1$

D. $\frac{1}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let ( $X, Y$ ) have joint probability density function

$$f(x,y) = \begin{cases} 8xy & \text{if } 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

If $E\left(X \mid Y = y_0\right) = \frac{1}{2}$, then $y_0$ equals

A. $\frac{3}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ have a binomial distribution with parameters $n$ and $p$, where $n$ is an integer greater than $1$ and $0 < p < 1$. If

$$P(X = 0) = P(X = 1),$$

then the value of $p$ is

A. $\dfrac{1}{n-1}$

B. $\dfrac{n}{n+1}$

C. $\dfrac{1}{n+1}$

D. $\dfrac{1}{n+1^{\frac{1}{n-1}}}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $\Phi(\cdot)$ denote the cumulative distribution function of a standard normal random variable. If the random variable $X$ has the cumulative distribution function

$$F(x) = \begin{cases} \Phi(x) & \text{if } x < -1 \\ \Phi(x+1) & \text{if } x \geq -1 \end{cases}$$

then which one of the following statements is true?

A. $P(X \leq -1) = \frac{1}{2}$

B. $P(X = -1) = \frac{1}{2}$

C. $P(X < -1) = \frac{1}{2}$

D. $P(X \leq 0) = \frac{1}{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Let $E$ and $F$ be two events such that $0 < P(E) < 1$ and $P(E \mid F) + P\left(E \mid F^c\right) = 1$. Then

A. $E$ and $F$ are mutually exclusive

B. $P\left(E^c \mid F\right) + P\left(E^c \mid F^c\right) = 1$

C. $E$ and $F$ are independent

D. $P(E \mid F) + P\left(E^c \mid F^c\right) = 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $A, B$, and $C$ be three independent events.
Define:
$\alpha = \Pr(A)$
$\beta = \Pr($ not all three events occur simultaneously $)$
$\gamma = \Pr($ at least one of the three events does not occur $)$
$\delta = \Pr(C$ occurs, but neither A nor B occur)
Then, the probability that $C$ occurs equals:

A. $\delta/(\beta + \delta)$

B. $\beta/(\beta + \delta)$

C. $\alpha\gamma(\beta + \delta)$

D. $\gamma/(\beta + \delta)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X_1 \sim U(0, 4)$, i.e., a uniform distribution on the interval $[0, 4]$.
Then, the value of

$$P\left(X_1 > 2\right) = ?$$

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $e^{-1}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let

$$X \sim \text{Binomial}(n, p)$$

and define the random variable

$$Z = n - X$$

Then the distribution of $Z$ is:

A. Binomial with parameters $(n, p)$
B. Binomial with parameters $(n, 1-p)$
C. Binomial with parameters $(n-1, p)$
D. Binomial with parameters $(n-1, 1-p)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If $A$ and $B$ are two events and the probability $P(B) \neq 1$, then evaluate the expression:

$$\frac{P(A) - P(A \cap B)}{1 - P(B)}$$

A. $P(A \mid \bar{B})$
B. $P(A \mid B)$
C. $P(\bar{A} \mid B)$
D. $P(\bar{A} \mid \bar{B})$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Suppose that the questions on a test are independent and identically distributed (i.i.d.), and you are able to answer any question correctly with probability $0.8$ . What is the expected number of questions you will attempt until you make your second incorrect answer?

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let $X, Y$ be random variables and $g : \mathbb{R} \to \mathbb{R}$. Assume all expectations exist. Which of the following statements are true?

A. $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ if $X$ and $Y$ are independent

B. $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$

C. $\mathbb{E}[Xg(Y) \mid Y] = g(Y)\mathbb{E}[X \mid Y]$

D. $\mathbb{E}[g(Y) \mid Y] = g(Y)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

For $X$ and $Y$, two jointly distributed random variables, which of the following statements is false?

A. If $X$ and $Y$ are independently distributed, then the conditional distribution of $Y$ given $X$ equals the marginal distribution of $Y$.

B. If the covariance between $X$ and $Y$ is $0$ , then $X$ and $Y$ are independent.

C. If covariance between $X$ and $Y$ is $0$ , then $X$ and $Y$ are not correlated.

D. If $X$ and $Y$ are independently distributed, then the joint density of $X$ and $Y$ is the product of their marginal densities.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Probability

The probability density function $f(x)$ of a random variable $X$ is symmetric about $0$ . Then

$$\int_{-2}^{2}\int_{-\infty}^{x} f(u)dudx = ?$$

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $E$ and $F$ be two independent events with

$$P(E \mid F) + P(F \mid E) = 1, P(E \cap F) = \frac{2}{9} \text{ and } P(F) < P(E)$$

Then $P(E)$ equals

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

The joint probability density function of the random variables $X, Y$ and $Z$ is

$$f(x,y,z) = 8xyz, \qquad 0 < x, y, z < 1 = \text{otherwise}$$

Then $P(X < Y < Z)$ is

A. $\frac{1}{8}$

B. $\frac{1}{3}$

C. $\frac{1}{6}$

D. $\frac{3}{8}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Which of the following functions is a probability density function of a random variable $X$ ?

A. $f(x) = \begin{cases} x(2-x), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$

B. $f(x) = \begin{cases} x(1-x), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$

C. $f(x) = \begin{cases} 2xe^{-x^2}, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$

D. $f(x) = \begin{cases} 2xe^{-x^2}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X$ and $Y$ be two continuous random variables with the joint probability density function

$$f(x,y) = \begin{cases} 2, & 0 < x + y < 1, x > 0, y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

$E\left(X \middle| Y = \frac{1}{2}\right)$ is

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $1$

D. $2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

An unbiased coin is tossed six times in a row and four different such trials are conducted. One trial implies six tosses of the coin. If $H$ stands for head ans $T$ stands for tail, the following are the observations from the four trials.

A. $HTHTHT$

B. $TTHHHT$

C. $HTTHHT$

D. $HHHT__$

Which statement describing the last two coin tosses of the fourth trial has the highest probability of being correct?

A. Two $T$ will occur.

B. One $H$ and one $T$ will occur.

C. Two $H$ will occur.

D. One $H$ will be followed by one $T$ .

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let a random variable $X$ follow a Poisson distribution such that

$$\text{Prob}(X = 1) = \text{Prob}(X = 2)$$

The value of $\text{Prob}(X = 3)$ is:

A. $\frac{4e^{-2}}{3}$

B. $\frac{9e^{-3}}{2}$

C. $\frac{2e^{-2}}{3}$

D. $\frac{3e^{-4}}{4}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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